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Subgrid combustion modeling of 3-D premixed flames in the thin-reaction-zone regime

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Abstract

Large eddy simulation (LES) is used to investigate three-dimensional (3D) lean premixed turbulent methane–air flames in the thin-reaction-zone regime. In this regime, the Kolmogorov scale is smaller than the preheat zone thickness, but larger than the reaction zone thickness. Past numerical studies of similar flames were primarily direct numerical simulation either in two-dimensions or using the artificially thickened flame approach in 3D. For an LES the effect of small (unresolved) scales on the scalar field must be modeled accurately to capture the correct flame structure. A subgrid combustion model based on the linear-eddy-mixing (LEM) model is used within an LES framework (called LEM–LES hereafter) to capture the 3D flame-structure of the highly stretched premixed flames. A finite-rate, one-step methane–air chemistry with a non-unity Lewis number formulation is used in this study. The simulated flame structure resembles flames experimentally studied in the thin-reaction-zone regime. Even though the preheat zone is broadened by the penetration of small eddies, the chemical reaction zone remains thin and localized. This feature is captured properly in the current LEM–LES approach. The flame structure and other statistics such as the flame area evolution, curvature, and strain-rate statistics computed using the LEM–LES are also in good agreement with the past DNS studies.

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Keywords: Premixed flames; Large-eddy simulation; Flame–turbulence; Subgrid model; Thin-reaction-zone

1. Introduction

Direct numerical simulation (DNS) has been the primary means to understand the physics of flame–turbulence interactions until recently [1,2]. However, since complete resolution of all length and timescales in reactive flows is prohibitively expensive, some simplifications are often used in these DNS studies. For example, 2D DNS studies using reduced but multi-step kinetics have been

carried out to understand the effect of stretch on flame structure and extinction in the flamelet regime [3,4]. On the other hand, 3D DNS studies [2] resort to very simple finite-rate chemistry and low Reynolds number.

Large eddy simulation (LES), on the other hand, has its own inherent constraint. In LES, all scales larger than the grid are resolved but the effect of scales smaller than the grid has to be modeled. However, for reacting flow typical flames are much smaller than the LES grid resolution, and even in the thin-reaction-zone (TRZ) regime where the flame preheat zone is broadened by turbulent eddies, the reaction zone is still very thin. Resolution of the reaction zone in 3D will

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require a grid resolution substantially larger than the grid required for resolving the turbulence scales. As a result, so far no LES study has attempted to resolve the true reaction zone thickness. Instead, past LES studies have used artificially thickened flame [5,6] or the G-equation [7] approach. In the artificially thickened flame approach, the chemical kinetic parameters are adjusted to artificially “thicken” the flame so that it can be resolved using a relatively coarse grid, whereas in the G-equation approach the flame is modeled as an infinitely thin front, and the structure is not resolved. Although both these approaches are powerful methods, they have their own limitations. For example, flame thickening diminishes the response of the flame to small scales, and the thin-flame approach requires a turbulent flame speed model that may not be valid in general [6].

In this study, a subgrid combustion model for LES based on the linear eddy mixing (LEM) model of Kerstein [8], is used to predict the 3D premixed flame structure in the TRZ regime. Earlier stand-alone application of the LEM model to flamelet and the TRZ regime using reduced multi-step kinetics [9,10] have demonstrated that this model is capable of capturing the changes in the flame structure between various regimes of burning without any specific adjustment. More recent application of LEM in LES [11,12] has demonstrated that the LEM model is capable of capturing flame structure in complex 3D swirling flows, not only in the TRZ regime but also in the broken-reaction-zone (BRZ) regime.

Figure 1 shows the the locations of the various flames investigated in the past (B flames, [9], F flames, [13]) in a regime diagram [14]. LM6000 and D2 are the premixed flames simulated in gas turbine combustors [11,12] using the LEM–LES

approach. The current flame in the TRZ regime is noted as TP in the figure and is similar to a past DNS flame [2]. The goal of the present effort was to demonstrate that the LEM–LES approach can capture flames in the TRZ regime without requiring excessive grid resolution and without resorting to any modification to the kinetics or the thickness of the flame. Comparison with past observations in both experiments and DNS will be used to demonstrate the overall accuracy of this LEM–LES approach in the TRZ regime.

2. Formulation and numerical implementation

The LES equations are obtained by Favre filtering the unsteady, compressible Navier–Stokes equations in generalized co-ordinates. Equations are not shown here for the sake of brevity and can be found elsewhere [15,11]. An additional equation for subgrid kinetic energy, k^{sgs} [15], together with an eddy viscosity assumption is used for the closure of the momentum and energy transport. A finite volume approach that is second-order accurate in space and time [11] is employed in the present study.

2.1. Scalar closure

In LEM–LES, molecular diffusion, small- and large-scale turbulent convection, and chemical reaction are all modeled separately but concurrently at their respective timescales. To briefly describe this model, split the velocity field as: $u_i = \tilde{u}_i + (u'_i)^R + (u'_i)^S$. Here, $\tilde{u}_i$ is the LES-resolved velocity field, $(u'_i)^R$ is the LES-resolved subgrid fluctuation (obtained from k^{sgs}), and $(u'_i)^S$ is the unresolved subgrid fluctuation. Then, consider the *exact* species equation for a scalar Y as:

$$\rho \frac{\partial Y}{\partial t} = -\rho [\tilde{u}_i + (u'_i)^R + (u'_i)^S] \frac{\partial Y}{\partial x_i} + \frac{\partial}{\partial x_i} (\rho Y V_{i,y}) + \rho \dot{\omega}_y \quad (1)$$

In LEM–LES, the above equation is rewritten as:

$$\frac{Y^* - Y^n}{\Delta t_{LES}} = -[\tilde{u}_i + (u'_i)^R] \frac{\partial Y^n}{\partial x_i}, \quad (2)$$

$$Y^{n+1} - Y^* = \int_t^{t+\Delta t_{LES}} \frac{1}{-\rho} \left[\rho (u'_i)^S \frac{\partial Y^n}{\partial x_i} + \frac{\partial}{\partial x_i} (\rho Y V_{i,y}) + \rho \dot{\omega}_y \right] dt'. \quad (3)$$

Here, Δt_{LES} is the LES time-step, ρ , V_m and $\dot{\omega}$ are, respectively, the mixture density, species diffusion velocity, and reaction rate. Eq. (2) describes the large-scale 3D LES-resolved convection of the scalar field and is implemented via Lagrangian

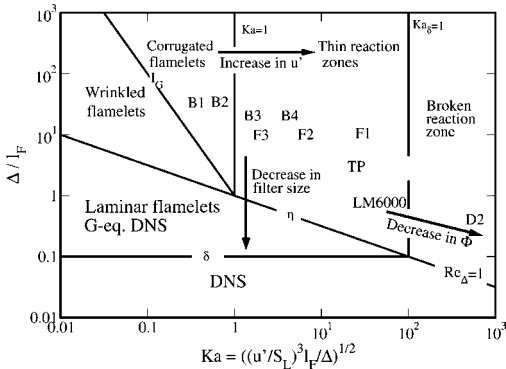


Fig. 1. Regime diagram for premixed combustion [14]. Here, Δ is the LES grid resolution. LM6000 is a General Electric combustor and D2 represents a DOE-HAT combustor near lean-blow-out. TP indicates the current simulated case.

transfer of mass across the finite-volume cell surfaces. Eq. (3) describes the subgrid LEM model as viewed on the LES space and timescale. The integrand includes all processes that occur below the LES grid, e.g., subgrid stirring, subgrid molecular diffusion, and reaction kinetics that occur within each LES cell. These processes are modeled on a 1D domain embedded inside each LES grid where the integrand is rewritten in terms of the subgrid time and space scales, so that:

$$\rho \frac{\partial Y^k}{\partial \tau^k} + F_s^k + \frac{\partial}{\partial s}(\rho Y^k V^k) = \rho \dot{\omega}^k. \quad (4)$$

A similar equation can be written for the subgrid energy [10]. Here, superscript k indicates that the subgrid field is further discretized within each LES by N_{LEM} number of subgrid cells along the local coordinate s such that the LES-resolved quantity Y is obtained by an ensemble average of the subgrid field. Thus, $Y = \frac{1}{N_{\text{LEM}}} \sum_{k=1}^{N_{\text{LEM}}} Y^k$. The 1D domain is aligned in the direction of the maximum scalar gradient [8], and the length of the LEM domain is equal to that of the local LES filter width, $\bar{\Delta}$.

In the above equation, subgrid stirring is symbolically represented as F_s^k , since it is implemented using stochastic re-arrangement events called *triplet maps* [8]. Each triplet map represents an *instantaneous* action of an isotropic turbulent eddy on the subgrid scalar field. The eddy size l is picked up randomly from an eddy size distribution $f(l)$ in the range $\bar{\Delta}$ to η (Kolmogorov scale), and stirring events occur at a specified frequency. Both $f(l)$ and stirring frequency are obtained using inertial-range scaling laws [8].

The LEM closure is similar to the closure in PDF methods [16], except that molecular diffusion is also included exactly in LEM–LES. As in PDF methods, the large-scale transport is modeled as a Lagrangian transport of the scalar fields across LES cells, and the subgrid stirring is modeled. In PDF methods, a mixing model is often employed to model turbulent mixing, whereas in LEM–LES, small-scale turbulent stirring is implemented by the triplet mapping process.

Conservation of mass, momentum, and energy (at the LES level), and conservation of mass, energy, and species (at the LEM level) are strongly coupled [17,11]. Chemical reaction at the LEM level determines heat release and thermal expansion at the LEM level, which at the LES level generates flow motion that, in turns, transports the species field at the LEM level. Full coupling is maintained in the LEM–LES to ensure local mass conservation.

2.2. Initial and boundary conditions

The earlier DNS [2] employed a global one-step mechanism ($R \rightarrow P$) with equal molecular

weights and a 129^3 grid resolution. Here, a 64^3 grid is used to resolve the same domain with 12 linear-eddy cells per LES cell. This implies that the LEM resolution, $\Delta X_{\text{LEM}} = \eta/2 = \delta_F/5$, where δ_F is the flame thickness defined as the ratio of thermal diffusivity to laminar flame speed S_L . Also, $(\eta/\delta_F = 0.27)$, which implies that the smaller turbulent scales can penetrate into the flame and influence the species and energy transport. However, the flame is thin compared to the integral scales ($L/\delta_F = 5.2$) and the turbulence is intense relative to the laminar flame speed ($u'/S_L = 10$). Thus, this flame is in the TRZ regime.

The present study uses a single-step, five-species (CH_4 , O_2 , CO_2 , H_2O , and N_2) methane–air mechanism with finite-rate kinetics [18], temperature dependent properties, and non-unity Lewis numbers (0.62–1.27). Stiffness in the chemistry is handled by using the DVODE package along with an operator splitting algorithm. This operator splitting between convection and chemistry limits the overall accuracy of the scheme to second-order.

The LES is initialized in the 3D domain with a plane laminar lean ($\phi = 0.8$) premixed flame (obtained using the PREMIX-CHEMKIN package) separating reactants (on the left, at $T = 570$ K and 1 atm) from the products (on the right). Characteristic based inflow–outflow boundary conditions [19] are employed in the streamwise direction, and periodic conditions are imposed in the transverse and spanwise directions. Turbulence is introduced at the inflow using a prescribed spectrum, as in the earlier DNS [2]. The overall incoming Mach number is around 0.013, and the Reynolds numbers based on the integral scale and the Taylor's scale are 70 and 50, respectively. For this Re, around 90% of the total kinetic energy, is resolved in the LES.

3. Results and discussion

Figure 2 shows the iso-surface of temperature ($T = 2000\text{K}$) together with the contours of vorticity magnitude at the inflow and the outflow plane. The interaction of the vorticity field with the flame front causes intense wrinkling and stretching. This wrinkling increases the flame surface area of the initially flat laminar flame. The suppression of the post flame velocity fluctuations due to the increased viscosity is evident, and the penetration of the small scales into the flame causes a broadening of the flame.

It is well known [2] that the stretching and wrinkling of the flame causes a dramatic increase in flame surface area. A progress-variable c (based on the methane mass fraction) surface is used to define the flame front location, where $c = 1 - Y_{\text{CH}_4}/Y_{\text{CH}_4,u}$. Here, subscript u indicates the value in the unburned mixture. A $c = 0.8$

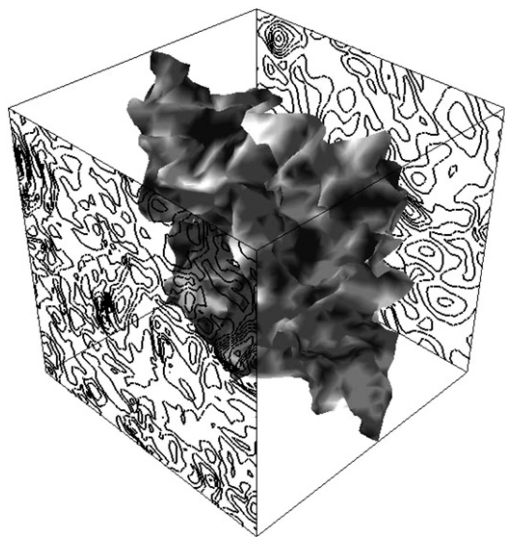


Fig. 2. Iso-surface of an instantaneous flame surface and contours of vorticity magnitude.

surface is used as the flame surface, as done earlier [2]. The flame surface area is computed as a function of time and is shown in Fig. 3. The flame surface area increases almost monotonously in time, even though the slope decreases eventually. The increase occurs as the turbulence wrinkles the initially planar flame. As turbulence decays, the rate of increase of the flame surface area decreases, and hence the slope of the plot also decreases. The current results are in reasonable agreement with the earlier DNS study [2] considering that there are differences between the two studies.

In the TRZ regime [20], $\delta_{RZ} < \eta < \delta_F$, where δ_{RZ} is the reaction-zone thickness. Hence, small eddies can penetrate into the preheat zone but not into the reaction zone. This is shown in Figs. 4 and 5 for two arbitrary time instants.

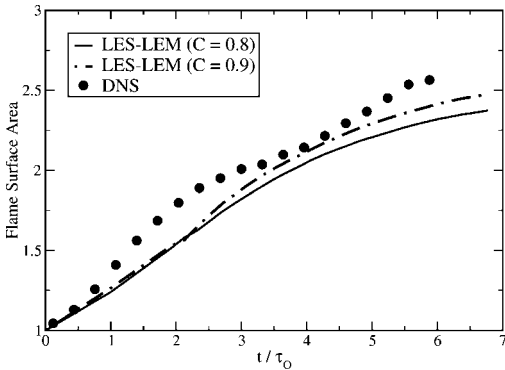


Fig. 3. Evolution of the flame surface area as a function of normalized time.

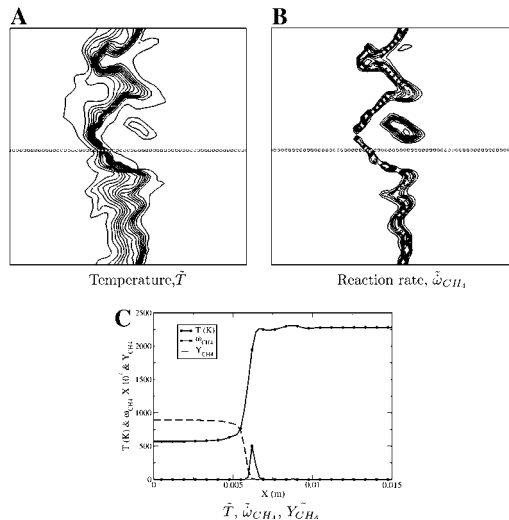


Fig. 4. Contours of LES-resolved temperature and methane reaction-rate at time, $t = T_1$. Dotted lines in (A) and (B) indicate locations where the line plots in (C) are obtained.

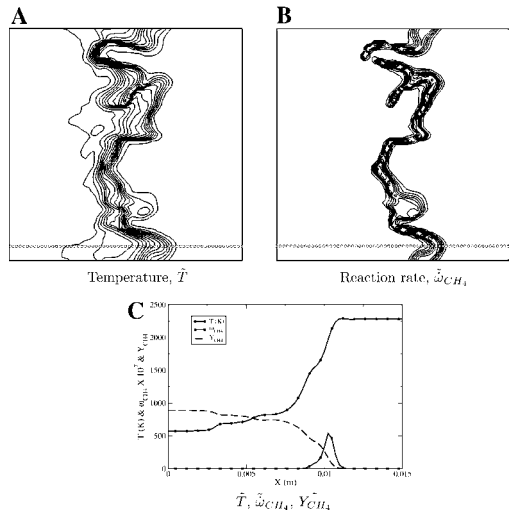


Fig. 5. Contours of LES-resolved temperature and methane reaction-rate at time, $t = T_2$. Dotted lines in (A) and (B) indicate locations where the line plots in (C) are obtained.

The instantaneous temperature contours (Figs. 4A and 5A) show the broadened preheat zone while the contours of methane reaction rate (Figs. 4B and 5B) reveal the localized nature of thin-reaction-zone.

The effect of preheat zone broadening is apparent in Fig. 5C whereas, in Fig. 4C the temperature and the $\hat{\omega}_{CH_4}$ profiles resemble the structure of a flame in the flamelet regime. Note that, all these

contours and profiles are spatially filtered (i.e., LES-resolved) values of the subgrid properties. Therefore, although the fields as shown here reveal only large-scale wrinkling, the effect of the small-scale wrinkling (due to the subgrid eddies) on the scalar field is included in the LEM–LES. This is a unique feature of the LEM–LES approach and is one of the major reasons why the flame structure in the TRZ regime can be captured.

The profile of methane mass fraction as shown in Fig. 5C suggests the presence of mixed partially burned and unburned fluids ahead of the reaction zone. This is also characteristic of flames in the TRZ regime, where small eddies transport heat and mass away from the reaction zone into the preheat zone.

3.1. Flame curvature

Flame curvature influences the local burning rate and hence, the propagation speed for flames with non-unity Lewis numbers. For $Le < 1$, the reactants diffuse more rapidly than thermal energy. Therefore, in convex (to the reactants) regions of the flame, the reactants have a focusing effect. This, in turn, increases the local reaction rate. The opposite situation occurs if the flame is concave to the reactants. For $Le > 1$, thermal energy is de-focused (focused) in convex (concave) regions of the flame, thus reducing (increasing) the flame propagation speed. This coupling of differential diffusion with the curvature has been studied extensively in strained laminar flames, and in both two-dimensional [4] and three-dimensional turbulent flames [1,2]. The current LES–LEM predicted curvature effects are discussed in this context.

Instantaneous reaction rate contours of methane (not shown, for brevity) showed that the elements of flame surface that are concave toward the reactants burn faster, while the elements that are concave toward the products have lower burn-

ing rates compared to a planar flame. This is confirmed in Fig. 6, which shows the reaction rate along two flame elements that are concave and convex to the reactants, respectively.

Typical PDF of flame curvature normalized by the laminar flame thickness and the PDF of the curvature shape factor are shown in Fig. 7. Curvature of the flame surface is given by $1/R = \nabla \cdot \mathbf{n}_{c=c_0}$, where $\mathbf{n} = -\nabla c/|\nabla c|$ points into the reactants. Curvature is computed from the following curvature tensor [21].

$$h_{ij} = -\frac{\partial^2 c}{\partial x_i \partial x_j} \frac{1}{g} + \frac{1}{g^3} \sum_{k=1}^3 \frac{\partial c}{\partial x_i} \frac{\partial c}{\partial x_k} \frac{\partial^2 c}{\partial x_j \partial x_k}, \quad (5)$$

where $g = |\nabla c|$. This tensor has two real eigenvalues h_1, h_2 and one zero eigenvalue. Using this tensor and its eigenvalues, the two principal curvatures of a surface are defined as $h_1 = 1/R_1$ and $h_2 = 1/R_2$. The surface curvature of a flame is then defined as the half-sum of the two local principal radii of curvature. Negative curvature represents a concave flame with respect to the reactants. The mean curvature normalized by flame thickness in Fig. 7 shows that the mean is zero, and the distribution is slightly skewed toward positive curvature. This is consistent with the results obtained in the past [1,4].

The curvature shape factor is a quantitative measure of the shape of the surface element and is defined as $S_h = h_s/h_l$, where h_s and h_l are the smaller and larger magnitudes of the eigenvalues of the curvature tensor h_{ij} . By definition, $S_h = +1$ implies a spherical surface element, $S_h = -1$ implies a pseudo-spherical (saddle-shaped) surface element, and $S_h = 0$ means that the element is cylindrical in shape. Figure 7 shows that the cylindrical curvature predominates in the flow. It is also clear that the spherical saddle points occur with a small but finite probability. These results are in excellent agreement with the earlier DNS studies [1,4].

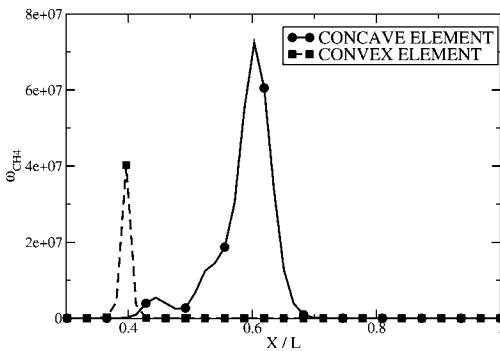


Fig. 6. Variation of methane reaction rate across convex and concave surfaces.

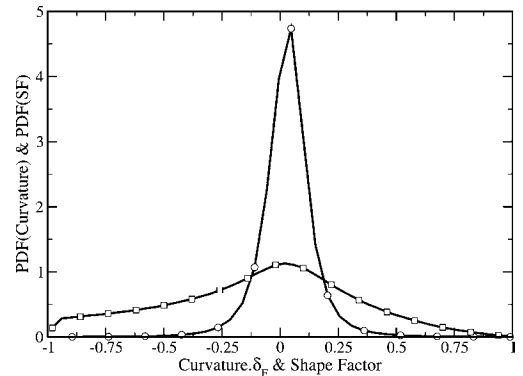


Fig. 7. PDF of normalized flame curvature and flame shape-factor. ○, curvature; □, shape factor.

3.2. Strain-rate and flame alignment

Strain-rate modifies the convective transport along the normal and the tangential directions to the flame, thus modifying the flame structure. Strain-rate also influences the evolution of the flame surface area and hence, alters the flame propagation speed. Figure 8 shows the pdf of the tangential strain-rate on the flame normalized by the Kolmogorov timescale. The tangential strain rate is computed as: $a_T = \partial u_k / \partial x_k - n_i S_{ij} n_j$. The PDF reveals that the mean tangential strain rate is positive and thus, the extensive strain-rate is dominant in the flame. This agrees with earlier DNS [1]. In the presence of heat release, flow acceleration normal to the flame is high and hence, the extensive (positive) strain-rate is high across the flame. As a result, the flame aligns preferentially with extensive strain-rate direction.

The strain-rate tensor has three real eigenvalues α , β , and γ , and three corresponding eigen-directions e_α , e_β , and e_γ . Past studies [22–24] in isotropic turbulence have shown that $\alpha > \beta > \gamma$ with $\alpha > 0$ and $\gamma < 0$ and β taking positive and negative values. Typically, α is associated with the extensive strain-rate, and $\gamma < 0$ is associated with the compressive strain-rate.

Figure 9 shows the alignment statistics between the flame normal and the strain-rate eigenvectors. The PDF is maximum for the cosine of the angle between the flame normal and the extensive strain-rate eigenvector e . This implies that the flame normal is most likely aligned with the positive, strain-rate and is in good agreement with the earlier observation in exothermic stagnation point flames [25].

It is shown earlier that the flame surface is predominantly cylindrical (zero mean curvature), and that the probability of the positive strain-rate is high on the flame surface. This implies that any other shaped surface such as a spherically curved surface (with a finite non-zero probability) is strained into a more cylindrical shape by the predominant extensive strain-rate. Thus, the regions

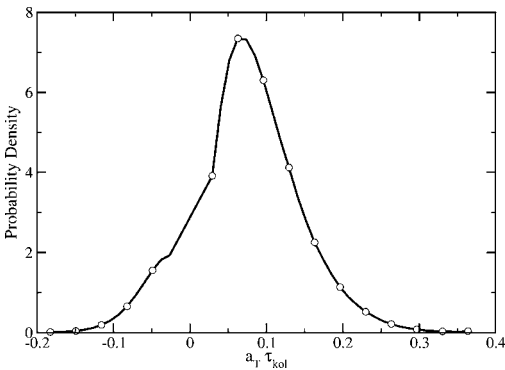


Fig. 8. PDF of the normalized tangential strain-rate.

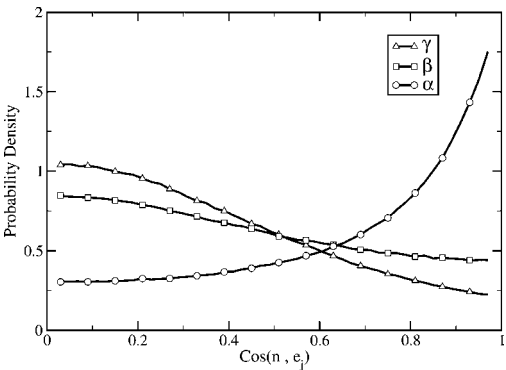


Fig. 9. Alignment of strain-rate eigenvectors with flame normal direction

of high (non-zero) curvature are possible only if the local strain-rate is low and vice versa. This notion is consistent with earlier observations [1], where they observed that the maximum strain is found in regions of zero curvature, while the maximum positive and negative curvature occurs where the strain is nearly zero.

Finally, the PDF of the eigenvalues of the strain-rate and the orientation of strain-rate eigen-directions relative to the vorticity field are shown in Figs. 10A and B, respectively. As seen earlier

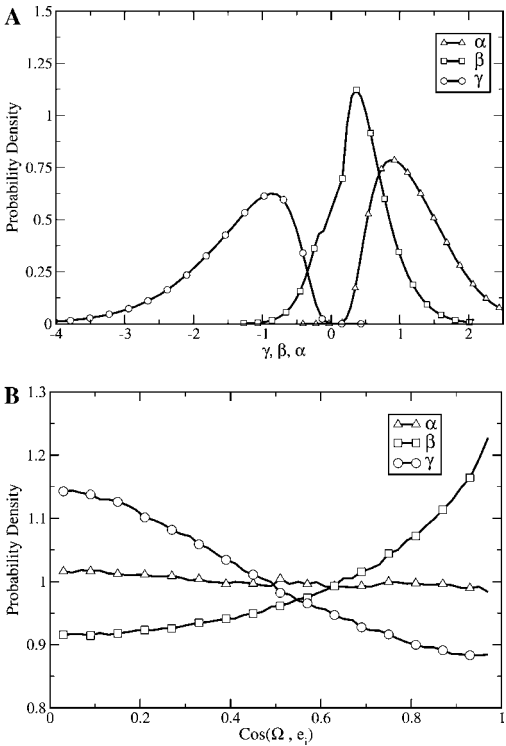


Fig. 10. PDF of Strain-rate eigenvalue and eigenvector alignment with vorticity.

[22–24], the intermediate strain-rate eigenvalue is predominantly positive and it is clear from Fig. 10B that the vorticity field is aligned preferentially with this intermediate eigenvalue.

In summary, the LES-resolved features extracted from the subgrid LEM–LES are in excellent agreement with past DNS studies primarily because the current approach captures the small-scale scalar features that exist below the LES resolution. Extension of the LEM–LES approach to deal with more complex chemistry is straightforward although the increase in computational cost for multi-step kinetics will remain a major hindrance. Methods such as in situ adaptive tabulation [26] have already been used in LEM [10] and may provide the necessary cost reduction for 3D LEM–LES of flames with detailed kinetics.

4. Conclusions

3D LES of freely propagating premixed flame in the TRZ regime is carried out. A single-step, 5-species finite-rate kinetics with variable Lewis number is used within a subgrid combustion model that explicitly simulates flame-turbulent interaction at all the scales. It is shown that the subgrid LEM model accounts for the small-scale wrinkling and hence, the flame structure consistent with the TRZ regime is properly captured. Many of the features, such as the broadening of the flame structure, the statistics of flame curvature, the flame-surface area evolution, the alignment of strain-rate tensor with the flame normal, and the vorticity field are in good agreement with the DNS studies. The current study confirms the potential of the LES-LEM approach for application to flames of practical interest in the TRZ regime.

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Comment

Jacqueline H. Chen, Sandia National Laboratories, USA. In the thin reactions zone regime curvature dissipation is important at the small scales. How does the linear eddy model capture multi-dimensional effects in the unresolved scales? Were corrections to this model implemented to account for non-local effects? Have you tested this model with larger filter sizes than a factor of two from the DNS?

Reply. In LEM, “exact” reaction-diffusion equations (for species and temperature in one dimension) are solved and hence it is not possible to include the curvature correction terms. In LEMLES, the effect of subgrid

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turbulence (including the effect of curvature due to unresolved subgrid scales) is modeled using 3D inertial range scaling laws. Using the definition of Menon and Kerstein [1], for the radius of curvature of the premixed flames in the 1D domain of LEM as half the distance between two adjacent flame crossings, Smith and Menon (1997), computed the PDF of the curvature and obtained very good comparisons with experiments in the flamelet regime. In the thin-reaction-zone regime simulated here, the broadened flame structure is actually resolved over multiple LES cells, and hence, most of the flame curvature is actually resolved on the LES grid and only the very fine-scale effects are modeled in the subgrid.

Yes, this LEMLES model has been used to simulate gas turbine combustor flows ([11,12] in paper) where the LES filter size is much larger than a factor of two from the DNS.

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